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ORIGINAL RESEARCH

CORONARY

Relative Fractional Flow Reserve Increase and Final Fractional Flow Reserve After Percutaneous Coronary Intervention

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ABSTRACT

BACKGROUND Post-percutaneous coronary intervention (PCI) fractional flow reserve (FFR) and relative FFR increase after PCI are determined by the interaction of baseline disease pattern, adequacy of PCI, and residual disease burden in a target vessel. However, the prognostic impact of relative FFR increase after PCI remains uncertain.

OBJECTIVES The aim of this study was to evaluate the prognostic relevance of relative FFR increase in addition to post-PCI FFR in patients undergoing PCI.

METHODS From the International Post PCI FFR Extended Registry, 1,497 patients with pre-PCI FFR ≤0.80 were analyzed. The relative FFR increase was expressed as a percentage and calculated as percentage FFR increase (ΔFFR%) with PCI ([post-PCI FFR − pre-PCI FFR]/pre-PCI FFR × 100). The primary endpoint was 5-year target vessel failure (TVF), a composite of cardiac death, target vessel myocardial infarction, or target vessel revascularization (TVR). TVR was further specified as target lesion revascularization (TLR) or non-TLR TVR.

RESULTS Maximally selected log-rank statistics identified 15.0% and 0.80 as the cutoffs to discriminate the occurrence of TVF for ΔFFR% and post-PCI FFR, respectively. Failure to meet either cutoff was associated with increased risk for TVF (ΔFFR%: 10.9% [122 of 1,148] vs 17.6% [59 of 349] [adjusted HR: 1.72; 95% CI: 1.23-2.42; $P = 0.002$]; post-PCI FFR: 10.9% [136 of 1,279] vs 21.3% [45 of 218] [adjusted HR: 2.08; 95% CI: 1.47-2.94; $P < 0.001$]). Patients with post-PCI FFR ≤0.80 had increased TVF compared with those with post-PCI FFR >0.80, regardless of sufficient (≥15.0%) or insufficient (<15.0%) relative FFR increase. The higher risk for TVF in patients with post-PCI FFR ≤ 0.80 compared with those with post-PCI FFR >0.80 was driven primarily by non-TLR TVR (2.2% vs 6.0%; $P = 0.015$) in the sufficient ΔFFR% group (≥15.0%), whereas it was attributable mainly to TLR (6.6% vs 17.0%; $P = 0.008$) in the insufficient ΔFFR% group (<15.0%).

CONCLUSIONS Relative FFR increase after PCI provides prognostic implications comparable with post-PCI FFR. Among patients with insufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with an increased risk for restenosis within stented segments, whereas among those with sufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with risk driven by disease progression in nonstented segments.
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The fundamental goal of percutaneous coronary intervention (PCI) is relieving myocardial ischemia, not simply alleviating local stenosis; therefore, ischemia-directed PCI has been a standard care for patients with symptomatic coronary artery stenosis.[1] Fractional flow reserve (FFR) has been used as a surrogate marker of interrogated vessel-related myocardial ischemia, and FFR-guided PCI has been supported by a Class 1A recommendation in the current guidelines.[2,3]

Beyond guiding treatment decisions, post-PCI FFR assessment has been evaluated, with previous studies demonstrating an inverse relationship between post-PCI FFR and adverse clinical events.[4,5] However, FFR reflects the cumulative hemodynamic impact of coronary atherosclerotic disease in the target vessel, and post-PCI FFR is determined by the interaction of baseline disease pattern, adequacy of PCI, and residual disease burden in a target vessel.[6] Therefore, post-PCI FFR alone cannot fully discriminate the relative contribution of each component. For example, even if local stenoses are optimally treated by PCI, post-PCI FFR could still be suboptimal because of residual diffuse disease in the nonstented segments. In this regard, evaluating the relative increase of FFR in combination with post-PCI FFR would help discriminate the relative contribution of stented and nonstented segment disease burden for risk assessment after PCI. Nevertheless, the prognostic impact of relative FFR increase after PCI remains uncertain.

The aim of the present study was to evaluate the prognostic implications of post-PCI FFR and relative increase of FFR after PCI from a multinational and multicenter individual patient-level registry of post-PCI FFR measurements.

METHODS

STUDY POPULATION. The present study population was derived from the extended follow-up data of the International Post PCI FFR Registry (**Figure 1**). The aim of the International Post PCI FFR Registry (NCT04012281) was to evaluate the predictive value of post-PCI FFR; it incorporated data from 4 registries across Korea, Japan, and China. Included registries were the COE-PERSPECTIVE (Influence of Fractional Flow Reserve [FFR] on the Clinical Outcomes of Percutaneous Coronary Intervention [PCI]: A Prospective, Multicenter FFR Registry; NCT01873560),[7] the DK CRUSH-VII (Post-DES FFR Predicts the Clinical Outcomes: DK CRUSH-VII, A Prospective, Multicenter, Registry Study; ChiCTR-PRCH-12001976),[8] the institutional registry of Tsuchiura Kyodo General Hospital in Ibaraki, Japan,[9] and the 3V-FFR-FRIENDS (Influence of Total Atherosclerotic Burden Assessed by 3-Vessel Fractional Flow Reserve [FFR] on the Clinical Outcomes of the Patients With Multi-Vessel Disease; NCT01621438) registry.[10] All patients included in the registry had undergone angiographically successful PCI with second-generation drug-eluting stents (DES) and post-PCI FFR measurement. Detailed information about this registry has been previously described.[11]

For the International Post PCI FFR Extended Registry, a standardized spreadsheet with uniform definitions of clinical outcomes was provided to the principal investigators of each registry. Of 2,200 patients initially enrolled in the International Post PCI FFR Registry, 2,128 patients were available for the 5-year follow-up data analysis. Among these patients, 526 patients who underwent PCI without pre-PCI FFR measurements and 105 patients with pre-PCI FFR >0.80 were excluded in the present analysis. In total, 1,497 patients who underwent PCI on the basis of pre-PCI FFR ≤0.80 were analyzed (**Figure 1**).

The study protocol was approved by the ethics committee at each participating center and adhered

ABBREVIATIONS AND ACRONYMS

DES = drug-eluting stent(s)

FFR = fractional flow reserve

ΔFFR% = percentage fractional flow reserve increase

MI = myocardial infarction

PCI = percutaneous coronary intervention

%DS = percentage diameter stenosis

TLR = target lesion revascularization

TVF = target vessel failure

TVR = target vessel revascularization

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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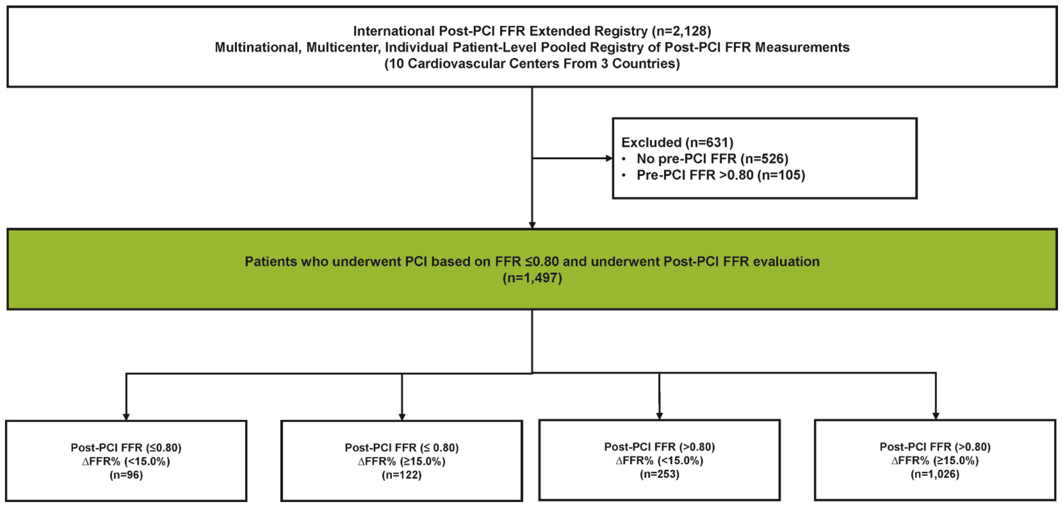
FIGURE 1 **Study Flow**



Study flow demonstrating the number of enrolled patients from registry. These patients were classified into 4 groups according to percentage fractional flow reserve increase (ΔFFR%) and post-percutaneous coronary intervention (PCI) fractional flow reserve (FFR) values.

to the principles of the Declaration of Helsinki. The requirement to obtain patient informed consent was waived because of the retrospective nature of the study. This study also complied with the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines.

CORONARY ANGIOGRAPHY, INTERVENTION, AND PHYSIOLOGICAL ASSESSMENT. Coronary angiography was conducted according to standard techniques. Quantitative coronary angiography was performed at designated core laboratories using a validated software program with optimal projections (CAAS II [Pie Medical Systems] and QAngio XA [Medis Medical Imaging]). Reference vessel diameter, post-PCI minimal luminal diameter, and post-PCI percentage diameter stenosis (%DS) were measured. PCI was performed following standard techniques with second-generation DES. The choices of stenting techniques, DES types, and the additional use of intracoronary imaging devices were at the discretion of physicians. Both pre- and post-PCI FFR measurements were obtained consecutively during the same index PCI procedure according to standardized protocols used in previous studies.[6,12] The present study used the final post-PCI FFR values from each procedure, which was measured at the same location as the pre-PCI measurement using the same pressure wire systems and hyperemic agents. For patients with multivessel interrogation, the target vessel was defined as the vessel with the lowest post-PCI FFR value. Percentage FFR increase (ΔFFR%) was defined as the relative change in FFR after PCI ([post-PCI FFR − pre-PCI FFR]/pre-PCI FFR × 100).

DEFINITIONS OF CLINICAL OUTCOMES. The primary clinical outcome was target vessel failure (TVF) at 5 years, defined as a composite of cardiac death, target vessel-related myocardial infarction (MI), or clinically driven target vessel revascularization (TVR). The secondary outcomes included a composite of cardiac death or target vessel-related MI, the individual outcomes of the primary outcome, target lesion revascularization (TLR), and non-TLR TVR at 5 years. The target lesion was defined as the stented segment, including the 5-mm proximal and distal margins. All deaths were considered cardiac deaths unless a noncardiac reason was indicated. MI was defined according to the fourth universal definition of MI,[13] and procedure-related MI was not considered a clinical event. Clinically driven revascularization was defined as repeat revascularization in the presence of ≥70% diameter stenosis or ≥50% diameter stenosis with at least 1 of the following: 1) recurrence of anginal symptoms; 2) positive noninvasive test results; or 3) positive invasive physiological test results. TVR was further specified as TLR or non-TLR TVR according to the recurring lesion location.

The target lesion included a 5-mm margin proximal and distal to the stent and the stent itself. Non-TLR TVR was a lesion in the target vessel that was outside of the target lesion by at least 5 mm distal or proximal to the target lesion. All instances of TLR and TVR were adjudicated through a post hoc analysis.

STATISTICAL ANALYSES. Categorical variables were expressed as numbers and relative frequencies (percent) and continuous variables as mean ± SD or medians (Q1-Q3), according to their distributions, which were checked using the Kolmogorov-Smirnov test. Student's *t*-test or the Wilcoxon signed rank test was performed for comparison of continuous variables according to the distributions of variables. The chi-square test for trend was used to evaluate categorical variables.

The optimal cutoff values of post-PCI FFR and ΔFFR% for predicting TVF at 5 years were calculated on the basis of maximally selected log-rank statistics[14] using the maxstat package in R. The cumulative incidence of clinical events was presented as Kaplan-Meier estimates and compared using a log-rank test or the Breslow test. Multivariable mixed-effects Cox proportional hazards regression was used to calculate the adjusted HR and 95% CI to compare the risk for clinical events between groups, classified according to optimal cutoff values of post-PCI FFR and ΔFFR%. Using the coxme package in R, the random effects for post-PCI FFR were incorporated at the participating centers to account for unmeasured heterogeneity across centers. The fixed-effect covariates with clinical relevance or univariate relationships with the outcome ($P < 0.10$) were entered into multivariable mixed-effects Cox proportional hazards regression model. Variables selected for inclusion were carefully chosen, given the number of events available, to ensure parsimony of the final models. Adjusted covariables were age, male, diabetes mellitus, dyslipidemia, acute coronary syndrome, left anterior descending coronary artery, location of lesion in the target vessel, pre-PCI reference diameter, pre-PCI diameter stenosis, pre-PCI lesion length, pre-PCI FFR, use of intravascular imaging, and participating centers. The assumption of proportionality was assessed graphically using a log-minus-log plot, and Cox proportional hazards models for all clinical outcomes satisfied the proportional hazards assumption. The assumption of proportional hazards model was evaluated using a 2-sided test of the scaled Schoenfeld residuals over time at a level of 0.05. Missing data were excluded from the analyses, and multiple imputation was not performed. All statistical analyses were 2-sided, with a significance threshold of $P < 0.05$. Statistical analyses were performed using R version 4.2.3 (R Foundation for Statistical Computing).

RESULTS

BASELINE CHARACTERISTICS. Table 1 shows the baseline characteristics of the study population and target lesions. Among 1,497 patients, 736 (49.2%) presented with chronic coronary syndrome. More than one-half of the target lesions were in proximal segments of the left main to left anterior descending coronary arteries. Mean %DS and median pre-PCI FFR were 64.4% ± 15.4% and 0.70 (Q1-Q3: 0.60-0.76), respectively. The optimal cutoff values of post-PCI FFR and ΔFFR% for the occurrence of TVF were 0.80 and 15%, respectively (Supplemental Figure 1). A total of 218 patients (14.6%) demonstrated post-PCI FFR ≤0.80, and 349 patients (23.3%) had ΔFFR% <15.0%.

Patients with ΔFFR% <15.0% had a higher prevalence of diabetes mellitus, lower pre-PCI %DS, less frequent use of intravascular imaging during PCI, but similar post-PCI %DS compared with those with ΔFFR% ≥15.0%. Patients with post-PCI FFR ≤0.80 had a higher prevalence of multivessel disease, smaller reference vessel diameter, and higher post-PCI %DS than those with post-PCI FFR >0.80 (Table 1). Among the 4 groups classified by both post-PCI FFR and ΔFFR%, patients with post-PCI FFR ≤0.80 and ΔFFR% <15.0% had the smallest reference vessel diameter and the highest post-PCI %DS (Supplemental Table 1).

CLINICAL OUTCOMES ACCORDING TO POST-PCI FFR AND ΔFFR%. Figure 2 and Table 2 summarize clinical outcomes according to ΔFFR% or post-PCI FFR. Failure to meet either cutoff was associated with an increased risk for TVF (ΔFFR% ≥15.0% vs <15%: 10.9% vs 17.6% [adjusted HR: 1.72; 95% CI: 1.23-2.42; $P = 0.002$]; post-PCI FFR >0.80 vs ≤0.80: 10.9% vs 21.3% [adjusted HR: 2.08; 95% CI: 1.47-2.94; $P < 0.001$]) (Figure 2). Higher risk for TVF was driven mainly by TVR. Patients with ΔFFR% <15.0% showed higher risk for TVR, due mainly to increased TLR, than those with ΔFFR% ≥15.0%. Conversely, patients with post-PCI FFR ≤0.80 showed elevated risk for both TLR and non-TLR TVR compared with those with post-PCI FFR >0.80 (Table 2, Supplemental Figure 2).

Additionally, these different clinical outcomes associated with ΔFFR% or post-PCI FFR were not

TABLE 1 Baseline Clinical, Lesion, and Procedural Characteristics According to ΔFFR% and Post-PCI FFR

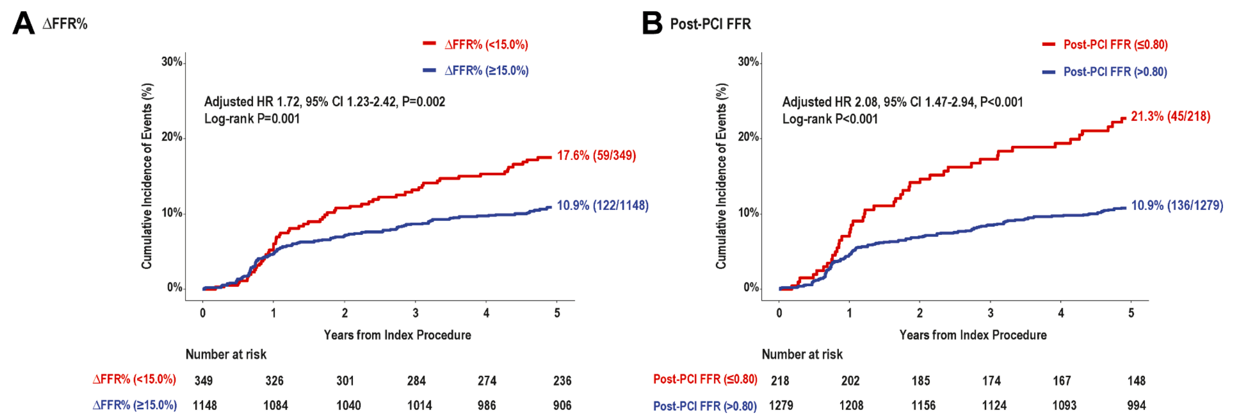
	ΔFFR% ≥15.0% (n = 1,148)	ΔFFR% <15.0% (n = 349)	*P* Value	Post-PCI FFR >0.80 (n = 1,279)	Post-PCI FFR ≤0.80 (n = 218)	*P* Value
Demographic characteristics						
Age, y	63.1 ± 10.0	64.2 ± 9.9	0.068	63.5 ± 9.9	62.4 ± 10.1	0.128
Male	888 (77.4)	275 (78.8)	0.621	985 (77.0)	178 (81.7)	0.152
Cardiovascular risk factors						
Hypertension	748 (65.2)	229 (65.6)	0.925	840 (65.7)	137 (62.8)	0.462
Diabetes mellitus	355 (30.9)	134 (38.4)	0.011	406 (31.7)	83 (38.1)	0.078
Hypercholesterolemia	542 (47.2)	198 (56.7)	0.002	617 (48.2)	123 (56.4)	0.031
Chronic renal failure	31 (2.7)	15 (4.3)	0.181	39 (3.0)	7 (3.2)	1.000
Current smoking	330 (28.7)	103 (29.5)	0.834	355 (27.8)	78 (35.8)	0.020
Previous MI	132 (11.5)	51 (14.6)	0.144	147 (11.5)	36 (16.5)	0.048
Previous PCI	243 (22.0)	88 (26.0)	0.136	281 (22.7)	50 (24.0)	0.741
Initial clinical manifestation						
Diagnosis			<0.001			<0.001
Silent ischemia	82 (7.6)	51 (14.6)		116 (9.1)	20 (9.2)	
Stable angina	350 (32.3)	148 (42.4)		426 (33.3)	90 (41.3)	
Unstable angina	493 (42.9)	91 (26.1)		526 (41.1)	58 (26.6)	
NSTEMI	91 (7.9)	25 (7.2)		93 (7.3)	23 (10.6)	
STEMI	49 (4.3)	12 (3.4)		58 (4.5)	3 (1.4)	
Others	62 (5.4)	22 (6.3)		60 (4.7)	24 (11.0)	
Acute coronary syndrome	633 (55.1)	128 (36.7)	<0.001	677 (52.9)	84 (38.5)	<0.001
LVEF	61.9 ± 8.1	61.8 ± 9.0	0.933	61.9 ± 8.4	61.8 ± 8.3	0.908
Lesion characteristics						
Multivessel disease	687 (59.9)	213 (61.0)	0.751	745 (58.3)	155 (71.1)	<0.001
Vessel			<0.001			0.003
Left main coronary artery/LAD	822 (71.6)	301 (86.2)		940 (73.5)	183 (83.9)	
LCx	127 (11.1)	17 (4.9)		134 (10.5)	10 (4.6)	
RCA	199 (17.3)	31 (8.9)		205 (16.0)	25 (11.5)	
Target lesion location			0.015			0.008
Proximal	631 (55.0)	170 (48.7)		664 (52.0)	137 (62.8)	
Mid	422 (36.8)	160 (45.8)		509 (39.8)	73 (33.5)	
Distal	81 (7.1)	17 (4.9)		90 (7.0)	8 (3.7)	
Unknown	13 (1.1)	2 (0.6)		15 (1.2)	0 (0)	
Angiographic characteristics						
Pre-PCI reference diameter, mm	2.9 ± 0.5	2.8 ± 0.5	<0.001	2.9 ± 0.5	2.7 ± 0.5	<0.001
Pre-PCI minimal lumen diameter, mm	1.0 ± 0.5	1.2 ± 0.4	<0.001	1.0 ± 0.5	1.0 ± 0.4	0.021
Pre-PCI diameter stenosis, %	66.8 ± 15.5	56.5 ± 12.0	<0.001	64.3 ± 15.3	65.2 ± 15.8	0.449
Pre-PCI lesion length, mm	25.7 ± 14.9	21.5 ± 11.8	<0.001	25.0 ± 14.5	22.9 ± 13.5	0.060
Post-PCI reference diameter, mm	3.0 ± 0.5	2.9 ± 0.4	0.001	3.0 ± 0.5	2.9 ± 0.5	<0.001
Post-PCI minimal luminal diameter, mm	2.7 ± 0.5	2.6 ± 0.4	<0.001	2.7 ± 0.5	2.6 ± 0.5	<0.001
Post-PCI diameter stenosis, %	9.7 ± 7.1	10.3 ± 7.0	0.174	9.6 ± 7.0	11.0 ± 7.7	0.006
Pre-PCI SYNTAX score	13.3 ± 7.5	13.3 ± 7.5	0.866	12.7 ± 7.1	16.9 ± 8.4	<0.001
Post-PCI SYNTAX score	3.6 ± 5.3	4.6 ± 6.1	0.009	3.3 ± 4.9	7.1 ± 7.3	<0.001
Procedural characteristics						
Mean number of implanted stents	1.5 ± 0.9	1.3 ± 0.6	<0.001	1.5 ± 0.9	1.3 ± 0.6	<0.001
Mean length of implanted stents, mm	34.3 ± 16.3	28.9 ± 13.0	<0.001	33.2 ± 15.7	32.4 ± 16.0	0.497
Pre-PCI FFR	0.65 ± 0.12	0.75 ± 0.04	<0.001	0.68 ± 0.11	0.63 ± 0.11	<0.001
Post-PCI FFR	0.89 ± 0.07	0.82 ± 0.05	<0.001	0.89 ± 0.05	0.76 ± 0.04	<0.001
ΔFFR%	43.9 ± 35.6	9.2 ± 3.8	<0.001	37.3 ± 34.5	26.9 ± 33.0	<0.001
Intravascular imaging			0.026			0.685
IVUS	531 (46.3)	130 (37.3)		564 (44.1)	97 (44.5)	
OCT	26 (2.3)	9 (2.6)		28 (2.2)	7 (3.2)	
Both	191 (16.6)	73 (20.9)		230 (18.0)	34 (15.6)	

Values are mean ± SD or n (%).

ΔFFR% = percentage fractional flow reserve increase; FFR = fractional flow reserve; IVUS = intravascular ultrasound; LAD = left anterior descending coronary artery; LCx = left circumflex coronary artery; LVEF = left ventricular ejection fraction; MI = myocardial infarction; NSTEMI = non-ST-segment elevation myocardial infarction; OCT = optical coherence tomography; PCI = percutaneous coronary intervention; RCA = right coronary artery; SYNTAX = Synergy Between PCI With TAXUS and Cardiac Surgery; STEMI = ST-segment elevation myocardial infarction.

demonstrated in patients who underwent intravascular imaging-guided PCI (ΔFFR% ≥15.0% vs <15%: 12.8% vs 18.2% [adjusted HR: 1.31; 95% CI: 0.79-2.19; *P* = 0.296]; post-PCI FFR >0.80 vs ≤0.80: 13.4% vs 16.6% [adjusted HR: 1.40; 95% CI: 0.80-2.46; *P* = 0.238]) compared with those who underwent angiography-guided PCI (ΔFFR% ≥15.0% vs <15%: 8.9% vs 16.7% [adjusted HR: 2.07; 95% CI: 1.17-3.67;

FIGURE 2 Target Vessel Failure According to ΔFFR% or Post-PCI FFR



Kaplan-Meier curves demonstrate the cumulative incidence of target vessel failure at 5 years after PCI according to ΔFFR% (A) and post-PCI FFR (B). Abbreviations as in Figure 1.

P = 0.013]; post-PCI FFR >0.80 vs ≤0.80: 8.3% vs 26.4% [adjusted HR: 3.82; 95% CI: 2.16-6.77; P < 0.001]) (Supplemental Figure 3).

PROGNOSTIC IMPLICATION OF THE COMBINATION OF POST-PCI FFR AND ΔFFR%. When the study population was classified according to combined patterns of post-PCI FFR and ΔFFR%, patients with post-PCI FFR ≤0.80 and ΔFFR% <15.0% had the highest incidence of TVF. Patients with either post-PCI FFR ≤0.80 or ΔFFR% <15.0% also had higher risk for TVF than those with post-PCI FFR >0.80 or ΔFFR% ≥15.0% (**Figure 3**, Supplemental Figure 4).

Patients with post-PCI FFR ≤0.80 demonstrated higher risk for TVF than those with post-PCI FFR >0.80, regardless of ΔFFR% (ΔFFR% ≥15.0%: 10.2% vs 16.6% [P = 0.014]; ΔFFR% <15.0%: 13.9% vs 27.6% [P = 0.007]) (**Table 3**). The higher risk for TVF in

TABLE 2 Comparison of Clinical Outcomes According to ΔFFR% or Post-PCI FFR

	ΔFFR% ≥15.0% (n = 1,148)	ΔFFR% <15.0% (n = 349)	Unadjusted HR (95% CI)	*P* Value	Adjusted HR (95% CI)[a]	*P* Value
TVF	122 (10.9)	59 (17.6)	1.65 (1.21-2.26)	0.002	1.72 (1.23-2.42)	0.002
Cardiac death	30 (2.7)	13 (3.9)	1.47 (0.77-2.82)	0.246	1.22 (0.61-2.44)	0.579
TVMI	11 (1.0)	3 (0.9)	0.91 (0.26-3.28)	0.891	0.82 (0.21-3.13)	0.771
TVR	94 (8.4)	46 (13.9)	1.66 (1.17-2.37)	0.005	1.91 (1.30-2.82)	<0.001
TLR	65 (5.8)	31 (9.4)	1.60 (1.05-2.46)	0.030	2.08 (1.30-3.35)	0.002
Non-TLR TVR	29 (2.6)	15 (4.5)	1.75 (0.94-3.27)	0.078	1.71 (0.87-3.34)	0.120
	Post-PCI FFR >0.80 (n = 1,279)	**Post-PCI FFR ≤0.80 (n = 218)**				
TVF	136 (10.9)	45 (21.3)	2.04 (1.46-2.86)	<0.001	2.08 (1.47-2.94)	<0.001
Cardiac death	33 (2.7)	10 (4.8)	1.81 (0.89-3.67)	0.101	1.82 (0.88-3.76)	0.105
TVMI	12 (1.0)	2 (0.9)	0.99 (0.22-4.43)	0.992	0.87 (0.20-4.01)	0.876
TVR	105 (8.4)	35 (16.9)	2.04 (1.39-2.99)	<0.001	2.09 (1.40-3.10)	<0.001
TLR	73 (5.9)	23 (11.1)	1.89 (1.18-3.02)	0.008	1.92 (1.17-3.13)	0.009
Non-TLR TVR	32 (2.6)	12 (5.9)	2.25 (1.16-4.37)	0.017	2.50 (1.26-4.97)	0.009

Values are n (%). [a]Multivariable analysis with a mixed-effects Cox proportional hazards regression with registry identifiers was performed on the basis of variables as follows: age, male, diabetes mellitus, dyslipidemia, acute coronary syndrome, left anterior descending coronary artery, location of lesion in target vessel, pre-PCI reference diameter, pre-PCI diameter stenosis, pre-PCI lesion length, pre-PCI FFR, intravascular imaging, and participating centers.

TLR = target lesion revascularization; TVF = target vessel failure; TVMI = target vessel-related myocardial infarction; TVR = target vessel revascularization; other abbreviations as in Table 1.

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TABLE 3 Comparison of Clinical Outcomes According to Post-PCI FFR, Stratified by ΔFFR%

	ΔFFR% ≥15.0%					ΔFFR% <15.0%				
	Post-PCI FFR >0.80 (n = 1,026)	Post-PCI FFR ≤0.80 (n = 122)	Unadjusted HR (95% CI)	Adjusted HR (95% CI)	*P* Value[a]	Post-PCI FFR >0.80 (n = 253)	Post-PCI FFR ≤0.80 (n = 96)	Unadjusted HR (95% CI)	Adjusted HR (95% CI)	*P* Value[a]
TVF	102 (10.2)	20 (16.6)	1.67 (1.04-2.70)	1.88 (1.14-3.10)	0.014	34 (13.9)	25 (27.6)	2.12 (1.27-3.56)	2.11 (1.23-3.62)	0.007
Cardiac death	25 (2.6)	5 (4.1)	1.67 (0.64-4.35)	2.50 (0.91-6.89)	0.076	8 (3.3)	5 (5.6)	1.71 (0.56-5.23)	2.39 (0.75-7.65)	0.142
TVMI	11 (1.1)	0 (0)	NA	NA	NA	1 (0.4)	2 (2.1)	5.40 (0.49-59.6)	8.75 (0.72-106.5)	0.089
TVR	79 (7.9)	15 (12.6)	1.62 (0.93-2.80)	1.75 (0.98-3.11)	0.058	26 (10.7)	20 (22.5)	2.21 (1.23-3.96)	2.04 (1.11-3.75)	0.021
TLR	57 (5.7)	8 (6.7)	1.17 (0.56-2.45)	1.33 (0.61-2.86)	0.472	16 (6.6)	15 (17.0)	2.68 (1.32-5.42)	2.52 (1.21-5.24)	0.013
Non-TLR TVR	22 (2.2)	7 (6.0)	2.69 (1.15-6.30)	2.93 (1.18-7.27)	0.020	10 (4.1)	5 (5.6)	1.36 (0.47-3.99)	1.17 (0.37-3.69)	0.783

Values are n (%). [a]Multivariable analysis with a mixed-effects Cox proportional hazards regression with registry identifiers was performed on the basis of variables as follows: age, male, diabetes mellitus, dyslipidemia, acute coronary syndrome, left anterior descending coronary artery, location of lesion in target vessel, pre-PCI reference diameter, pre-PCI diameter stenosis, pre-PCI lesion length, pre-PCI FFR, intravascular imaging, and participating centers.

Abbreviations as in **Tables 1 and 2**.

patients with post-PCI FFR ≤0.80 compared with those with post-PCI FFR >0.80 was driven primarily by non-TLR TVR (2.2% vs 6.0%; *P* = 0.020) in the sufficient ΔFFR% group (≥15.0%), whereas it was attributable mainly to TLR (6.6% vs 17.0%; *P* = 0.013) in the insufficient ΔFFR% group (<15.0%) (**Figure 4**).

When patients who met the cutoff of either post-PCI FFR or ΔFFR% were compared regarding the clinical outcomes, there was no significant difference in the risk for TVF and its individual components between the 2 groups (Supplemental Figure 5, Supplemental Table 2).

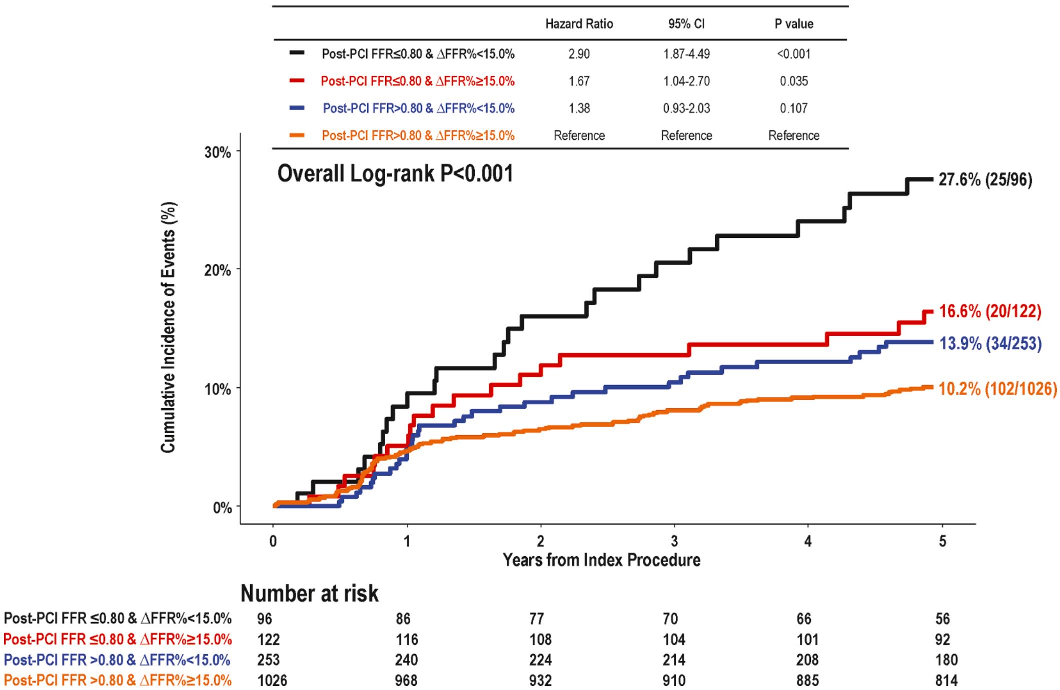
DISCUSSION

This study evaluated the clinical relevance of post-PCI physiological assessment using prospective, international, and long-term follow-up data (**Central Illustration**). The main findings are as follows. First, the best cutoff values for predicting cardiovascular events were 0.80 for post-PCI FFR and 15% for relative FFR increase at 5-year follow-up. Second, both post-PCI FFR ≤0.80 and ΔFFR% <15% were significantly associated with increased risk for TVF. Third, when patients were stratified according to post-PCI FFR and ΔFFR%, those with both post-PCI FFR ≤0.80 and ΔFFR% <15% showed the highest risk. Fourth, patients who achieved sufficient ΔFFR% but still had post-PCI FFR ≤0.80 had higher risk for disease progression in nonstented segments, whereas those with both post-PCI FFR ≤0.80 and insufficient ΔFFR% were predominantly at risk for restenosis within the stented segments.

Although numerous clinical trials support the clinical application of physiological assessment,[15,16] those studies have not assessed the physiological results after revascularization. In this regard, Hwang et al[5] reported individual patient-level pooled data, and post-PCI FFR was significantly associated with both TVF and cardiac death or MI, especially in patients with post-PCI FFR ≤0.80. These findings suggest that FFR-guided treatment decision is not the same as physiologically optimized PCI, and the physiological results after revascularization are significantly related to clinical outcomes. Although FFR reflects cumulative hemodynamic impact in the target vessel, PCI with stent implantation is a local treatment. Therefore, physiological results after PCI are influenced by underlying disease patterns,[1] especially diffuse residual atherosclerosis in the nonstented segment,[17] a masked second lesion beyond the stented segment, or suboptimal expansion or other acute procedural complications in the stented segment.[17-19] However, post-PCI FFR alone cannot discriminate the potential domain of physiologically suboptimal results after PCI between stented and nonstented segments. The relative increase in FFR reflects the adequacy of PCI, and information about relative increase of FFR in combination with post-PCI FFR would be helpful to discriminate the relative contributions of stented and nonstented segments for risk assessment after PCI.

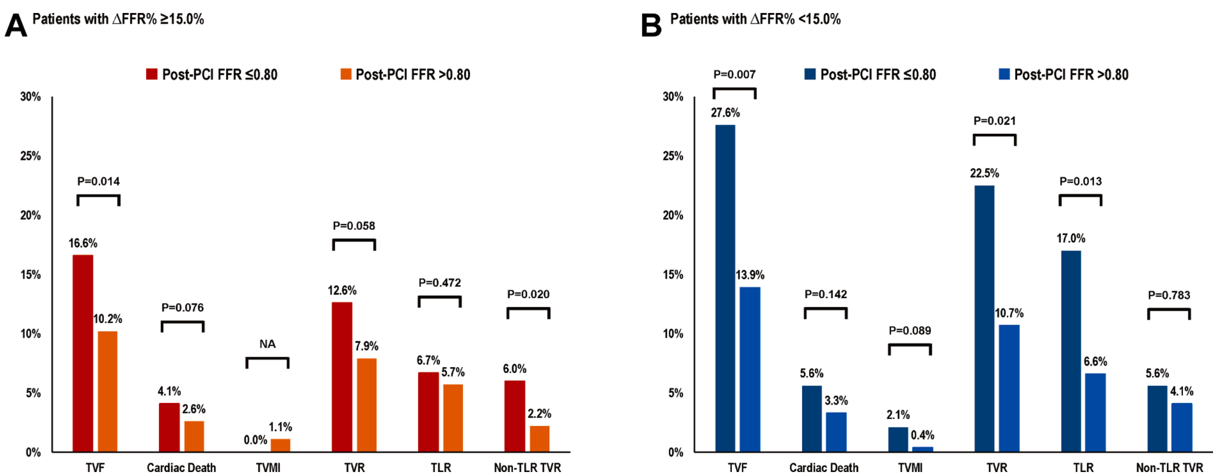
Previously, Lee et al[6] evaluated 621 patients who underwent FFR-guided PCI and had available post-PCI FFR values. In that study, both low post-PCI FFR (<0.85) and low ΔFFR% (<15%) were associated with increased risk for adverse events at 2-year follow-up. The present analysis revalidated the prognostic impact of post-PCI FFR and ΔFFR% using international, prospective, and long-term follow-up data. Considering that regional differences exist in clinical patterns in daily practice, the current

FIGURE 3 Target Vessel Failure According to Patterns by ΔFFR% and Post-PCI FFR



Kaplan-Meier curves demonstrate the incidence of target vessel failure at 5 years after PCI according to patterns by ΔFFR% and post-PCI FFR. Abbreviations as in Figure 1.

FIGURE 4 Comparison of Clinical Outcomes According to Post-PCI FFR, Stratified by ΔFFR%



Bar graphs demonstrate the cumulative incidence of primary and secondary endpoints according to post-PCI FFR among patients with sufficient ΔFFR% (≥15.0%) (A) and patients with insufficient ΔFFR% (<15.0%) (B) after PCI. TLR = target lesion revascularization; TVF = target vessel failure; TVMI = target vessel-related myocardial infarction; TVR = target vessel revascularization; other abbreviations as in Figure 1.

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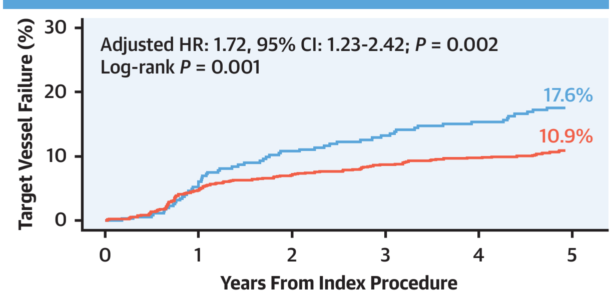
CENTRAL ILLUSTRATION Outcomes According to Relative FFR Increase and Final FFR After PCI

Clinical Relevance of Post-PCI Physiologic Assessment by Relative FFR Increase and Final FFR

A Study Population

- Post-PCI FFR Extended registry from Korea, Japan, China
- 2,200 patients initially enrolled; 1,497 patients underwent PCI based on FFR ≤0.80 and post-PCI FFR evaluation, and were available for 5-year follow-up
- Among 1,497 patients, 736 (49.2%) presented with chronic coronary syndrome
- Mean percent diameter stenosis was 64.4% ± 15.4% and median pre-PCI FFR was 0.70 (Q1-Q3: 0.60-0.76)

B Relative FFR Increase Predicting Target Vessel Failure

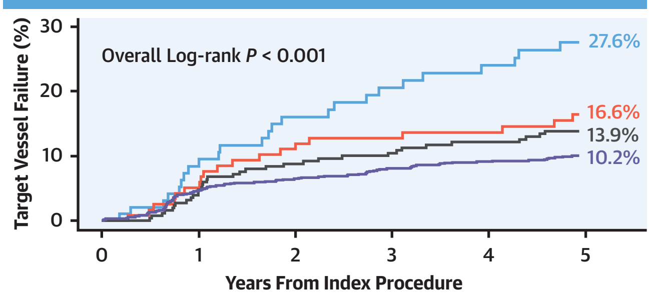


No. at Risk:

349	326	301	284	274	236
1,148	1,084	1,040	1,014	986	906

ΔFFR% (<15.0%)
ΔFFR% (≥15.0%)

C TVF According to Patterns by Percent FFR Increase and Post-PCI FFR



No. at Risk:

96	86	77	70	66	56
122	116	108	104	101	92
253	240	224	214	208	180
1,026	968	932	910	885	814

Post-PCI FFR ≤0.80 and ΔFFR% <15.0%
Post-PCI FFR ≤0.80 and ΔFFR% ≥15.0%
Post-PCI FFR >0.80 and ΔFFR% <15.0%
Post-PCI FFR >0.80 and ΔFFR% ≥15.0%

- Relative FFR increase after PCI provides prognostic implications comparable to post-PCI FFR.
- Among patients with insufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with an increased risk of restenosis within stented segments.
- Among patients with sufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with risk driven by disease progression in nonstented segments.
- Relative FFR increase would provide complementary information to understand the mechanism behind suboptimal physiologic results after PCI.

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The relative increase in fractional flow reserve (FFR) provides prognostic implications comparable with post-percutaneous coronary intervention (PCI) FFR. Among patients with insufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with an increased risk for restenosis within stented segments, whereas among those with sufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with risk driven by disease progression in nonstented segments. These findings suggest that relative FFR increase can provide complementary information to understand the mechanism behind suboptimal physiological results after PCI. ΔFFR% = percentage fractional flow reserve increase.

multicenter and multinational study may be more generalizable across various clinical settings. Both post-PCI FFR and relative FFR increase were significantly associated with clinical outcomes at 5-year follow-up after PCI. In the present study, patients with both post-PCI FFR ≤0.80 and insufficient ΔFFR% <15% showed approximately 3-fold increased risk for TVF compared with those with post-PCI FFR >0.80 and sufficient ΔFFR%. Increased risk for TVF was due mainly to higher rates of TVR. However, the pattern of increased risk for TVR was different according to post-PCI FFR and ΔFFR%. Post-PCI FFR ≤0.80 was associated with increased risk for restenosis within the stented segments, leading to more frequent TLR, among those with relative FFR increases <15%. Conversely, post-PCI FFR ≤0.80 was associated mainly with the risk from disease progression in nonstented segments among those with relative FFR increases ≥15%. Considering that both the distribution of atherosclerosis (focal vs diffuse) and local severity (major FFR gradient vs no major FFR gradient) determine the pattern of coronary artery disease, post-PCI physiological assessment should evaluate both post-PCI FFR for cumulative hemodynamic impact in the target vessel and ΔFFR% for the adequacy of PCI in the stented segment.

Debating issues on the post-PCI physiological assessment is the optimal cutoff value of post-PCI FFR. The optimal cutoff value for predicant 5-year TVF was 0.80 in this study. It has been shown that cutoff values of post-PCI FFR ranged widely from 0.84 to 0.96 in the literature.[19] These differences may have stemmed from differences in the study population's characteristics, including target vessel location, treatment modality, or medical treatment after PCI. However, considering that the fundamental purpose of PCI is relieving myocardial ischemia, the optimal cutoff value for post-PCI FFR should be at least 0.80. Additionally, post-PCI FFR alone has limitations, as it cannot stratify the potential reasons for suboptimal post-PCI FFR, especially residual diffuse disease, which cannot be modified by PCI. To distinguish physiological outcomes and determine whether they can be improved with additional procedures, the integration of ΔFFR% and post-PCI FFR serves as a gatekeeper for further intervention. For example, for patients with post-PCI FFR ≤0.80, if their ΔFFR% was <15%, additional procedures could have improved clinical outcomes. Conversely, if ΔFFR% was satisfactory, further procedures would not have been beneficial for improving physiological results, represented by post-PCI FFR. Although trans-stent FFR gradient and instantaneous FFR gradient per unit time after PCI provide additional information on the physiological results of PCI,[20,21] those indexes require pull back tracing and additional calculation. The FFR pull back tracing curve offers lesion-specific details, while ΔFFR% provides vessel-specific insights to assess PCI adequacy in daily practice. Therefore, combined with post-PCI FFR pull back tracing, ΔFFR% may provide additional information about hemodynamic gain from local treatment.

When insufficient ΔFFR% or suboptimal post-PCI FFR is observed after PCI, several strategies may be considered. First, a comprehensive evaluation of the treated vessel, including the target lesion, can be performed using intravascular imaging modalities. Assessment of minimal luminal area, edge dissection, stent malapposition, or residual plaque burden by intravascular ultrasound or optical coherence tomography may identify suboptimally treated segments or angiographically hidden stenosis in nontarget lesion segments. After identification of these factors, targeted interventions may be undertaken, which could potentially improve ΔFFR% or post-PCI FFR and translate into better clinical outcomes. If low ΔFFR% or post-PCI FFR persists despite optimization efforts, operators may reasonably consider terminating further intervention once either ΔFFR% or post-PCI FFR reaches an acceptable threshold. Conversely, when both parameters remain suboptimal, intensified post-PCI surveillance during follow-up may be warranted to facilitate early reintervention or consideration of alternative revascularization strategies, such as coronary artery bypass grafting.

STUDY LIMITATIONS. First, because this was a post hoc retrospective analysis of ΔFFR%, several inherent limitations should be acknowledged. As the registry was not originally designed to investigate the prognostic implications of ΔFFR%, residual confounding factors cannot be fully excluded.

Second, because this was a patient-level analysis that could not demonstrate lesion-specific characteristics, pull back data of FFR were not collected, and further analysis according to lesion length was not feasible. However, the relationship between the pull back pressure gradient and clinical outcomes remains unclear, and lesions with high or low pull back pressure gradient demonstrated comparable prognosis in a previous study.[22]

Third, there were no mandated protocols for further optimization of the stented segment in cases of low post-PCI FFR. However, post-PCI FFR was measured after clinically and angiographically optimal results of PCI.

Fourth, intravascular imaging-guided procedural optimization was performed in about 45% of the study population. Therefore, the reason for insufficient ΔFFR% or suboptimal post-PCI FFR could not be fully assessed.

Fifth, the treating physician was not blinded to the post-PCI FFR values. Nevertheless, ΔFFR% was not a predefined endpoint.

Sixth, although the prevalence of acute coronary syndrome was higher in the optimized result groups (ΔFFR% ≥15% or post-PCI FFR >0.80) by chance, clinical outcomes were significantly better than those in the suboptimized result groups even after multivariable adjustment. As coronary flow reserve and the index of microcirculatory resistance were not systematically measured, the potential influence of combined microvascular disease on post-PCI FFR or

ΔFFR%, especially in patients with acute coronary syndrome, could not be addressed.

Seventh, significant differences in the risk for TVF was driven mainly by the risk for TVR. However, TVR events were objectively adjudicated according to predefined criteria.

CONCLUSIONS

Both relative FFR increase and post-PCI FFR showed similar prognostic implications after PCI during 5-year follow-up. Among patients with insufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with an increased risk for restenosis within stented segments, whereas among those with sufficient relative FFR increase, post-PCI FFR ≤0.80 was associated with risk driven by disease progression in nonstented segments. These findings suggest that relative FFR increase would provide complementary information to understand the mechanism behind suboptimal physiological results after PCI.

DATA SHARING STATEMENT Anonymized patient-level data will be made available by the corresponding author for reasonable requests. Consent was not obtained for data sharing, but the presented data are anonymized, and risk for identification is minimal.

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PERSPECTIVES

WHAT IS KNOWN? Post-PCI FFR and ΔFFR% after PCI are determined by the interaction of baseline disease pattern, adequacy of PCI, and residual disease burden in the target vessel. Nevertheless, the prognostic impact of ΔFFR% after PCI remains uncertain.

WHAT IS NEW? The best cutoff values for predicting cardiovascular events were 0.80 for post-PCI FFR and 15% for relative FFR increase at 5-year follow-up. Both post-PCI FFR ≤0.80 and ΔFFR% <15% were significantly associated with increased risk for TVF at 5 years. When patients were stratified according to post-PCI FFR and ΔFFR%, those with both post-PCI FFR ≤0.80 and ΔFFR% <15% showed the highest risk. Patients who achieved sufficient ΔFFR% (≥15%) but still had post-PCI FFR ≤0.80 had higher risk for disease progression in nonstented segments, whereas those with both post-PCI FFR ≤0.80 and insufficient ΔFFR% (<15%) were predominantly at risk for restenosis within the stented segments.

WHAT IS NEXT? These findings suggest that ΔFFR% after PCI can provide complementary information to understand the mechanism behind suboptimal physiological results after PCI.

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KEY WORDS clinical outcome, coronary artery disease, fractional flow reserve, percutaneous coronary intervention, prognosis

APPENDIX For supplemental figures and tables, please see the online version of this paper.